

Module 2

Chapter 5

Projection

It's often necessary to figure out "how much" a vector points along a certain direction or directions. To do that, we use vector projections. The word “project” hints at shadows cast on the wall on which a projector light is shining. A *projection* tries to capture *a vector with a representation that uses fewer dimensions*.

Projecting from *three dimensions to two* looks like dropping a *shadow onto the two-dimensional surface*; the “shadow” is the projection (Fig 2), and the shadow and the original vector form a *right triangle* (Fig1). A projection from **two**

dimensions to one also looks like a *right triangle*. From higher dimensions, we could project down to three or two dimensions to try to visualize the data. **In all**

cases, the vector between the projection and the original is orthogonal to the surface it is projected on. Projections can be useful for making approximations to

data that fit a simplified, low-dimensional model.

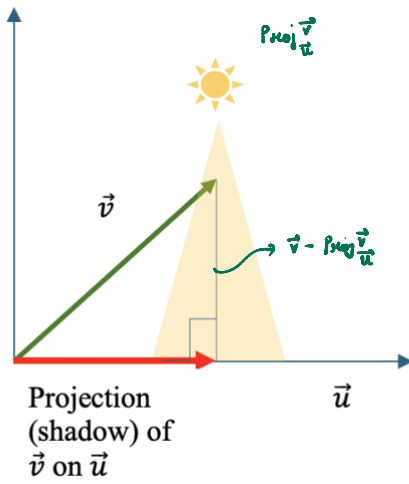


Fig 1

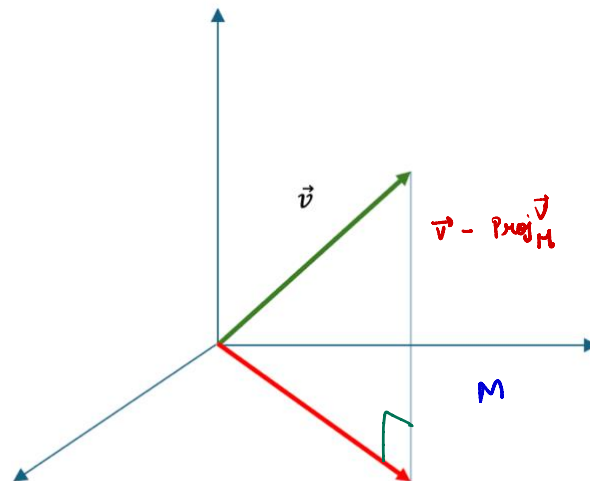


Fig 2

Vector projection of one vector onto another

If we have two vectors $\vec{v} = (v_1, v_2)$ and $\vec{u} = (u_1, u_2)$ and we want to project $\vec{u}$ on $\vec{v}$.

We write it as: **$\text{Proj}_{\vec{v}} \vec{u}$** and it represents the red vector in Fig 3 and is given by:

$$\text{Proj}_{\vec{v}} \vec{u} = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \vec{v} = \frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v}$$

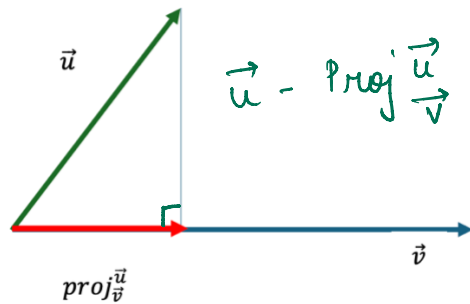


Fig 3

How?

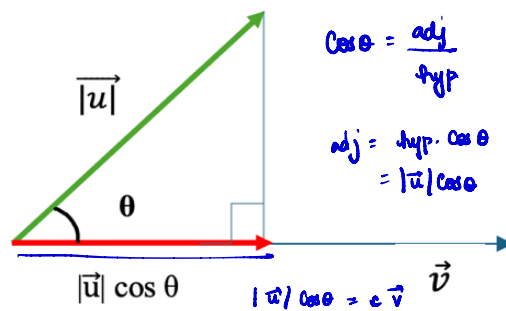
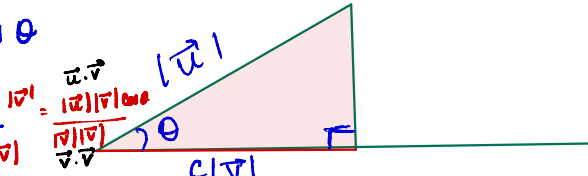


Fig 4

$$\cos \theta = \frac{c|\vec{v}|}{|\vec{u}|}$$

$$c|\vec{v}| = |\vec{u}| \cos \theta$$

$$c = \frac{|\vec{u}| \cos \theta}{|\vec{v}|}$$



Let's assume the line has been given to us in the form of some vector $\vec{v}$ that is in the correct direction; the line is made up of all possible multiples of $\vec{v}$, including negative multiples. We want the multiple that represents our projection, which means that it forms the base of a right triangle that sits on the line (Fig 4).

Suppose we're looking for the projection of $\vec{u}$ onto the line that is the span of $\vec{v}$.

Using high school trigonometry, we know the base of this right triangle, the

multiple of $\vec{v}$ we want, has a length of $|\vec{u}| \cos \theta$. So,

$$c|\vec{v}| = |\vec{u}| \cos \theta,$$

for some constant c that we're looking for to find the projection. If we try to solve for c , we get

$$c = \frac{|\vec{u}| \cos \theta}{|\vec{v}|}$$

Multiplying the top and bottom by $|\vec{v}|$, we get:

$$c = \frac{|\vec{u}||\vec{v}| \cos \theta}{|\vec{v}||\vec{v}|} = \frac{|\vec{u}||\vec{v}| \cos \theta}{|\vec{v}|^2 \cos 0^\circ} = \frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}}$$

We can now express the scaling factor of $\vec{v}$ as a ratio of two dot products. The final projection vector, then, is:

$$\text{Proj}_{\vec{v}} \vec{u} = c \cdot \vec{v} = \frac{\vec{u} \cdot \vec{v}}{|\vec{v}|^2} \vec{v} = \frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v}$$

Let's look at an example:

What will be the projection of vector $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ on to vector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$?

Let $\vec{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$\text{proj}_{\vec{v}} \vec{u} = \frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \cdot \vec{v}$$

$$\text{Then, } \text{Proj}_{\vec{v}} \vec{u} = \frac{\begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}}{\begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ \frac{3}{2} \end{bmatrix}$$

A *practical application* of the above concept can be used in video games:

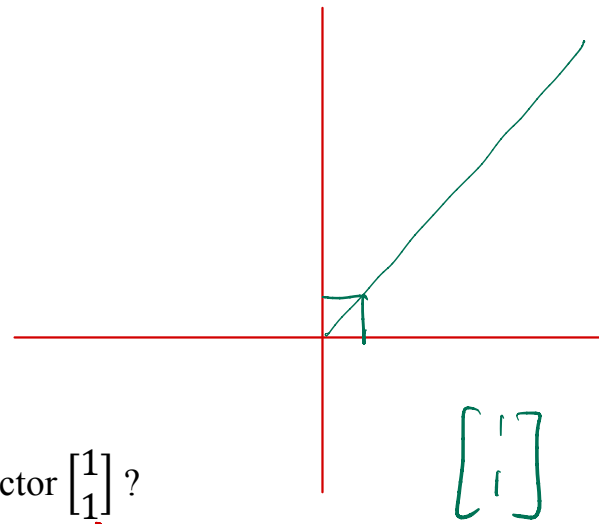
If a player was running with a velocity vector $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and ran into a wall at a 45

degree angle, the most natural response will be for the game to have the player

move along the wall, but not at full velocity. The new velocity vector would

generally be the component of the velocity that travels along the wall, which would

$$\text{be } \begin{bmatrix} \frac{3}{2} \\ \frac{3}{2} \end{bmatrix}.$$



Let's Practice:

Find the projection of $\vec{u} = (4,3)$ onto $\vec{v} = (2,8)$.
Find the vector component of $\vec{u}$ orthogonal to $\vec{v}$

Projection in higher dimensions:

We may want to project down onto a plane instead of a line, or generally project a vector in n dimensions down to a **subspace of $m < n$** dimensions. This has applications in fitting linear models to data, as we'll see shortly, as well as applications in graphics (e.g. converting 3D models to 2D scenes) and machine learning (trying to represent data with fewer numbers, achieving compression and perhaps reducing noise... the possibilities!!!).

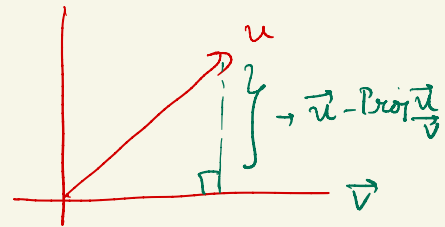
Let's now understand how to project vectors from a higher dimension space to a lower space in lower dimension. For visualization, we will be projecting a vector in a 3D space onto a 2D plane (Fig 5)

$$\text{ii) } \vec{u} = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} 2 \\ 8 \end{bmatrix}$$

$$\text{Proj}_{\vec{v}} \vec{u} = \frac{\begin{bmatrix} 4 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 8 \end{bmatrix}}{\begin{bmatrix} 2 \\ 8 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 8 \end{bmatrix}} \cdot \begin{bmatrix} 2 \\ 8 \end{bmatrix}$$

$$= \frac{8 + 24}{4 + 64} \begin{bmatrix} 2 \\ 8 \end{bmatrix} = \frac{32}{68} \begin{bmatrix} 2 \\ 8 \end{bmatrix} = \frac{8}{17} \begin{bmatrix} 2 \\ 8 \end{bmatrix}$$

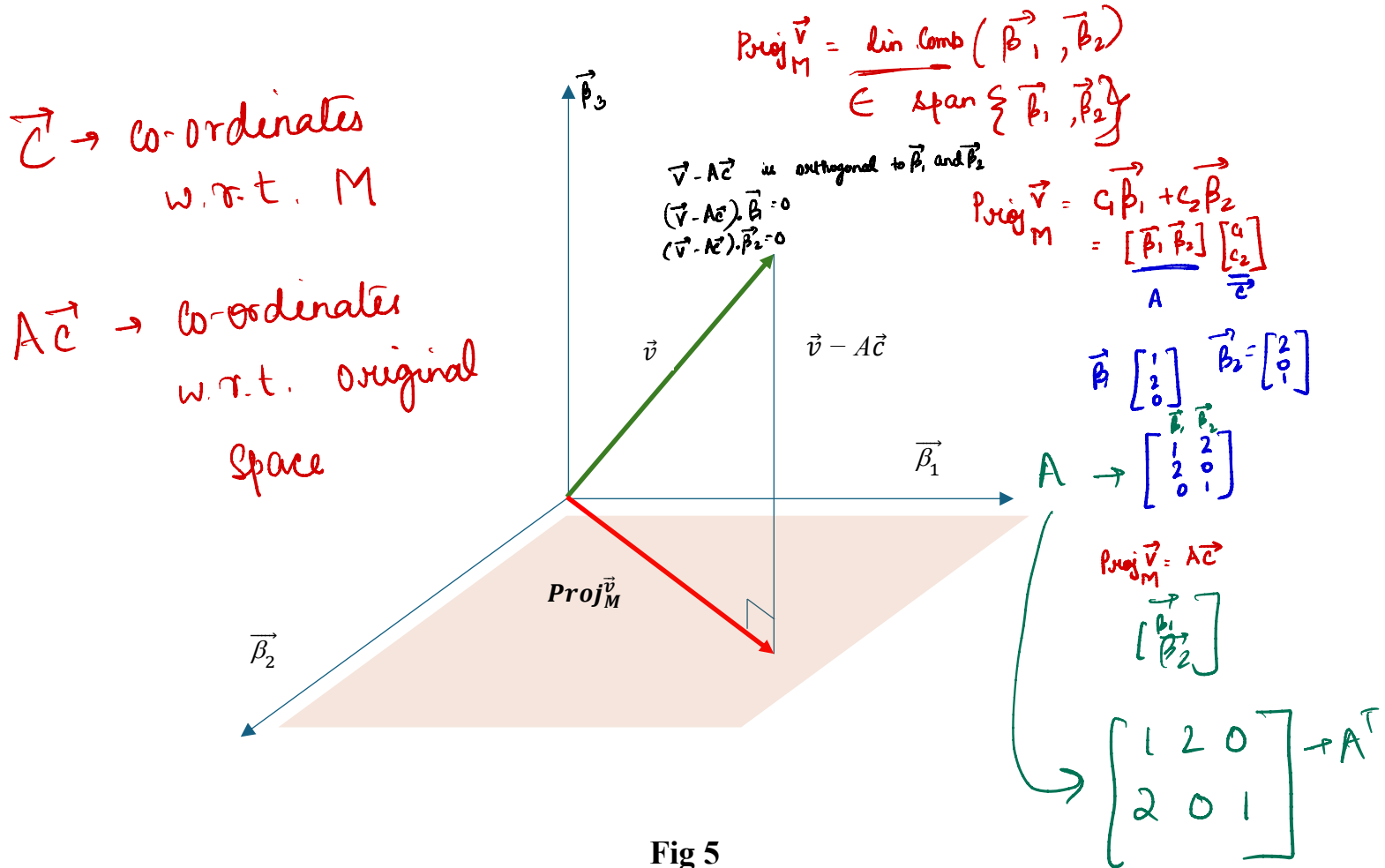
$$= \begin{bmatrix} 16/17 \\ 64/17 \end{bmatrix}$$



ii) Component of $\vec{u}$ orthogonal to $\vec{v}$

$$\text{Component of } \vec{u} \text{ orthogonal to } \vec{v} = \vec{u} - \text{Proj}_{\vec{v}} \vec{u}$$

$$= \begin{bmatrix} 4 \\ 3 \end{bmatrix} - \begin{bmatrix} 16/17 \\ 64/17 \end{bmatrix} = \begin{bmatrix} 52/17 \\ -13/17 \end{bmatrix}$$



Let $\vec{v}$ be a higher dimensional vector that we are projecting on a subspace M . The formula here is more complex than the formula for the line, but it will be

derived from two key facts:

First, the result of the projection must **belong to the subspace**, which means it must have some coordinates according to the basis spanning the subspace (linear combination of the basis vectors). Let $\vec{\beta}_1, \vec{\beta}_2, \dots, \vec{\beta}_n$ be the **basis vectors** spanning M . Then,

$$\text{Proj}_M \vec{v} = c_1 \vec{\beta}_1 + c_2 \vec{\beta}_2 + \dots + c_n \vec{\beta}_n$$

Let A be a matrix whose column vectors are basis vectors for M . We can then write the above as:

$$\text{Proj}_{\vec{M}} \vec{v} = A\vec{c}$$

Where $\vec{c}$ is a vector containing the scalars $c_1, c_2, \dots, c_n$

$$A = [\vec{\beta}_1 \vec{\beta}_2 \dots \vec{\beta}_n]$$

Second, we know that this vector $\vec{v} - A\vec{c}$ is *orthogonal* to every basis vector in A ,

and this results in a series of dot products equaling zero:

$$\vec{\beta}_1 \bullet (\vec{v} - A\vec{c}) = 0$$

$$\vec{\beta}_2 \bullet (\vec{v} - A\vec{c}) = 0$$

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$$\vec{\beta}_n \bullet (\vec{v} - A\vec{c}) = 0$$

$$\begin{bmatrix} \vec{\beta}_1 \\ \vec{\beta}_2 \\ \vdots \\ \vec{\beta}_n \end{bmatrix} \bullet (\vec{v} - A\vec{c}) = 0$$

$\underbrace{\hspace{10em}}_{A^T}$

This is a system of Linear equations that can be rewritten as:

$$A^T(\vec{v} - A\vec{c}) = 0$$

We will now use algebra to solve for $\vec{c}$, keeping in mind that we cannot divide by a matrix, but we can multiply by inverses:

- Distributive property: $A^T\vec{v} - A^TA\vec{c} = 0$

- Add to both sides: $A^T\vec{v} = \underbrace{(A^TA)^{-1}}_{(A^TA)^{-1}} \underbrace{A^TA}_{A^TA} \vec{c}$

- Multiply both sides by $(A^TA)^{-1}$ on the left: $(A^TA)^{-1}A^T\vec{v} = \{(A^TA)^{-1}A^TA\} \vec{c}$

- Use definition of Matrix inverse: $\underbrace{(A^TA)^{-1}A^T}_{\text{identity}} \vec{v} = \mathbf{I}\vec{c}$

$$\vec{c} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

- Use property of Identity Matrices: $(A^T A)^{-1} A^T \vec{v} = \vec{c}$
- $\vec{c}$ is the vector of coefficients $c_1, c_2, \dots, c_n$

We know:

$$\text{Proj}_{\vec{M}} \vec{v} = A \vec{c}$$

Replacing the value of $\vec{c}$:

$$\text{Proj}_{\vec{M}} \vec{v} = A(A^T A)^{-1} A^T \vec{v}$$

Let's take an example:

Suppose we are projecting 3d vectors onto the plane, say M, that is the span of $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

and $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. Matrix A will be : $\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$

$$A^T = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

Notice that A is not square, which is typical; a projection generally goes to fewer dimensions, making A “tall” because it has fewer basis vectors than the number of elements in the vectors.

$$A^T A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

The inverse also generally exists if the columns of $\mathbf{A}^T \mathbf{A}$ are linearly independent.

So,

$$(\mathbf{A}^T \mathbf{A})^{-1} = \begin{bmatrix} 2/3 & -1/3 \\ -1/3 & 2/3 \end{bmatrix}$$

$$(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T = \begin{bmatrix} 2/3 & 1/3 & -1/3 \\ -1/3 & 1/3 & 2/3 \end{bmatrix}$$

Suppose we want to project a vector $\vec{v} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ onto M and we want the co-ordinates

of the projected vector in the basis of M, given by the columns of A, We then

multiply $(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$ to $\vec{v}$:

$$(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \vec{v} = \begin{bmatrix} 2/3 & 1/3 & -1/3 \\ -1/3 & 1/3 & 2/3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \rightarrow \vec{c}$$

If we want to get the co-ordinates of the projected vector in the basis of the original space $\vec{v}$ was in, we multiply A to our result.

If we multiply by A on the left, we get back the original vector, because $\vec{v}$ was already on the plane to project onto (i.e. $\vec{v}$ was a linear combination of the basis vectors of M)! We wouldn't get back $\vec{v}$ if it was not a linear combination of the basis vectors of M.

Let's Practice

Find the projection of $(1,1,1)$ onto the space spanned by $\{(1,2,3), (4,5,6)\}$. Give the co-ordinates of the projected vector in terms of the basis vectors of the space on which we are projecting.

Also, verify if $\vec{v}$ is in the space spanned by $\{(1,2,3), (4,5,6)\}$

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \quad \vec{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} = \begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix}$$

$$(A^T A)^{-1} \rightarrow \det \begin{bmatrix} 14 & 32 \\ 32 & 77 \end{bmatrix} = 54$$

$$(A^T A)^{-1} = \frac{1}{54} \begin{bmatrix} 77 & -32 \\ -32 & 14 \end{bmatrix}$$

$$(A^T A)^{-1} A^T \vec{v} = \frac{1}{54} \begin{bmatrix} 77 & -32 \\ -32 & 14 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{54} \begin{bmatrix} 77 & -32 \\ -32 & 14 \end{bmatrix} \begin{bmatrix} 6 \\ 15 \end{bmatrix}$$

$$= \begin{bmatrix} -1/3 \\ 1/3 \end{bmatrix}$$

→ coordinates of
projected vector in terms
of the basis vectors of
the space we are projecting
on.

$A\vec{c}$